

SI26_P07: Sports Squad Constraint Checker

AI-Assisted Coding Interview Problem

Problem Statement

Build a compact Sports Squad Constraint Checker for students organizing an inter-hostel futsal match. It should validate a manually selected seven-player squad against a fixed local roster and show every squad rule that is satisfied or violated.

Use one attractive primary screen or report with the local roster, selection controls, a Validate Squad action, position and cohort counts, a rule summary, a validation message, and sample/reset controls. You may use a browser, desktop or mobile tools, a spreadsheet or notebook, or a CLI that produces a clear visual or tabular report. The tool validates the user's selection only: it must not optimize, select, rank, or recommend a squad, and it does not manage matches, tournaments, accounts, or bookings.

Load this local roster and select S01 through S07 as the built-in squad:

Player ID	Student	Position	Cohort	Availability	Selected
S01	Aditi Rao	GOALKEEPER	YEAR_2	AVAILABLE	Yes
S02	Bilal Khan	DEFENDER	YEAR_2	AVAILABLE	Yes
S03	Chitra Nair	DEFENDER	YEAR_3	AVAILABLE	Yes
S04	Deepak Shah	FORWARD	YEAR_2	AVAILABLE	Yes
S05	Esha Roy	FORWARD	YEAR_3	AVAILABLE	Yes
S06	Farhan Das	UTILITY	YEAR_2	AVAILABLE	Yes
S07	Gita Menon	UTILITY	YEAR_3	AVAILABLE	Yes
S08	Harish Patel	FORWARD	YEAR_2	UNAVAILABLE	No
S09	Imani Joseph	GOALKEEPER	YEAR_3	AVAILABLE	No

The built-in selection is valid: it has seven players, one goalkeeper, two defenders, two forwards, four YEAR_2 players, three YEAR_3 players, and no unavailable player. As the required one-change invalid demonstration, replace S07 with S08. The seven-player size and position rules still pass, but the result must list exactly PLAYER_UNAVAILABLE: S08 and COHORT_LIMIT_EXCEEDED: YEAR_2 has 5, maximum 4.

Contracts

- Each fixed roster row has one unique non-blank player ID, one position from GOALKEEPER, DEFENDER, FORWARD, or UTILITY, one cohort from YEAR_2 or YEAR_3, and availability AVAILABLE or UNAVAILABLE. Preserve roster order in the selection display.
- Treat the selected player IDs as roster references, and require them to be distinct. An unknown or repeated selected ID reports INVALID_SELECTION_REFERENCE and clears earlier rule results and counts.
- A valid squad has exactly 7 distinct selected players, exactly 1 selected goalkeeper, at least 2 selected defenders, at least 2 selected forwards, no selected unavailable player, and no more than 4 selected players from either cohort. UTILITY players count toward squad size and cohort totals but not defender or forward minimums.
- Evaluate every squad rule independently and list every violation. Use this order: SQUAD_SIZE_MUST_BE_7, GOALKEEPER_COUNT_MUST_BE_1, MINIMUM_DEFENDERS_NOT_MET, MINIMUM_FORWARDS_NOT_MET, one PLAYER_UNAVAILABLE: <ID> per unavailable selection in roster order, then one COHORT_LIMIT_EXCEEDED: <cohort> has <count>, maximum 4 per exceeded cohort in YEAR_2, YEAR_3 order.
- Show VALID only when there are no violations; otherwise show INVALID with the complete ordered list. Counts and individual rule states must always describe the current selection, not a previous validation.
- Reset restores the fixed roster and the built-in S01 through S07 selection. The checker must report rules only and must never generate or recommend a replacement squad.

Acceptance criteria

- **Required:** Load the built-in selection in one action and show VALID, squad size 7, position counts 1 goalkeeper, 2 defenders, 2 forwards, and 2 utility players, plus cohort counts 4 for YEAR_2 and 3 for YEAR_3.
- **Required:** Replace only S07 with S08; show INVALID with exactly PLAYER_UNAVAILABLE: S08 followed by COHORT_LIMIT_EXCEEDED: YEAR_2 has 5, maximum 4, while the size, goalkeeper, defender, and forward checks remain satisfied.
- **Required:** Reset, then deselect only S07; show squad size 6 and exactly SQUAD_SIZE_MUST_BE_7 with no additional violation.
- **Required:** Reset again and restore the original valid counts and rule states without retaining any earlier invalid message.
- **Required:** Keep the roster selection, counts, ordered violations, overall status, and sample/reset actions synchronized, and include focused checks for the valid baseline, the S07-to-S08 replacement, the six-player case, and the cohort limit of exactly 4.

- **Optional:** Add a compact visual formation view driven by the same selected-player and position data.

Use AI coding assistants. Before implementation, create a short plan with 3–5 ordered steps and useful checkpoints. Be prepared to present that plan, explain any changes you made to it, share relevant prompts, summarize your design, and show test evidence such as tests, screenshots, or output samples.

How You'll Be Evaluated

- **Planning and Solution Presentation:** Present your 3–5-step implementation plan, explain how the work followed or changed that plan, and demonstrate the working solution with clear explanations
- **AI Prompting Strategy:** Show the prompts you used to translate this problem statement into technical specifications for AI assistants
- **Design Constraints and Technology Choices:** Explain the constraints you provided to AI regarding design patterns, technology stack, and architectural decisions
- **AI-Influenced Decision Making:** Discuss trade-offs, assumptions, and how AI recommendations influenced your choices for components, data structures, and implementation approaches
- **Testing and Validation:** Demonstrate how you tested the application covering both typical usage scenarios and edge cases
- **Live Modification Capability:** Be prepared to implement one small modification, and possibly a second if time permits, using AI assistance; keep your development environment ready for focused changes and verification